Supplementary material: Optimal feedback control model details

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1 Methods

1.1 Linear-quadratic-Gaussian control

The state of the environment $\boldsymbol{x}_t$ evolves as a function of the previous state and the action $\boldsymbol{u}_t$ performed by the agent according to a discrete-time linear dynamical system with Gaussian noise

$$\boldsymbol{x}_{t+1} \sim \mathcal{N}(\mathbf{A}\boldsymbol{x}_t + \mathbf{B}\boldsymbol{u}_t, \boldsymbol{\Sigma}_{\boldsymbol{x}}), \tag{1}$$

whose covariance is $\boldsymbol{\Sigma}_{\boldsymbol{x}}$.

At each time step, the agent receives a noisy observation $\boldsymbol{y}_t$ of the state,

$$\boldsymbol{y}_t \sim \mathcal{N}(\mathbf{H}\boldsymbol{x}_t, \boldsymbol{\Sigma}_{\boldsymbol{y}}), \tag{2}$$

which is a linearly transformed version of the state with additive Gaussian noise with covariance $\boldsymbol{\Sigma}_{\boldsymbol{y}}$.

The cost function is quadratic in the state and the actions

$$\sum_{t=1}^{T} \boldsymbol{x}_t^\top \mathbf{Q} \boldsymbol{x}_t + \boldsymbol{u}_t^\top \mathbf{R} \boldsymbol{u}_t. \tag{3}$$

This control problem is called linear-quadratic-Gaussian (LQG) control and has a well-known optimal solution, which is composed of optimal state estimation (Kalman filter, KF Kalman, 1960a) and optimal control (linear-quadratic regulator, LQR Kalman, 1960b).

The KF iteratively computes a posterior distribution about the state given a sequence of sensory observations $p(\boldsymbol{x}_t|\boldsymbol{y}_1, \ldots, \boldsymbol{y}_t) = \mathcal{N}(\hat{\boldsymbol{x}}_t, \boldsymbol{\Sigma}_t)$, where the mean state estimate and its covariance are updated according to

$$\hat{\boldsymbol{x}}_{t+1} = \mathbf{A}\hat{\boldsymbol{x}}_t + \mathbf{B}\boldsymbol{u}_t + \mathbf{K}_{t+1}\left(\boldsymbol{y}_{t+1} - \mathbf{H}\left(\mathbf{A}\hat{\boldsymbol{x}}_t + \mathbf{B}\boldsymbol{u}_t\right)\right) \tag{4}$$

$$\boldsymbol{\Sigma}_{t+1} = (\mathbf{I} - \mathbf{K}_{t+1}\mathbf{H})\mathbf{P}_{t+1}, \tag{5}$$

where $\mathbf{K}_t$ is defined as

$$\mathbf{K}_{t+1} = \mathbf{P}_{t+1}\mathbf{H}^\top \left(\mathbf{H}\mathbf{P}_{t+1}\mathbf{H}^\top + \boldsymbol{\Sigma}_{\boldsymbol{y}}\right)^{-1}, \tag{6}$$

$$\mathbf{P}_{t+1} = \mathbf{A}\boldsymbol{\Sigma}_t\mathbf{A} + \boldsymbol{\Sigma}_{\boldsymbol{x}}. \tag{7}$$

$$\tag{8}$$

The linear-quadratic regulator defines an policy as a function of the current state estimate

$$\boldsymbol{u}_t = \mathbf{L}_t\hat{\boldsymbol{x}}_t, \tag{9}$$

which minimizes the expected cost by solving a dynamic programming problem. For a derivation of $\mathbf{L}_t$ and a more detailed derivation of $\mathbf{K}_t$, see e.g. Kochenderfer et al. (2022).

1.2 Tracking task with delayed observations

1.2.1 Velocity control model

In the velocity control model, the latent state contains the target position and the subject's hand position,

$$\boldsymbol{x}_t = \begin{bmatrix} x_t^{\text{right}} \\ x_t^{\text{left}} \end{bmatrix}, \tag{10}$$

where the target follows a random walk and the subject controls the velocity of the left hand. The generative dynamics are

$$\boldsymbol{x}_{t+1} = \mathbf{A}\boldsymbol{x}_t + \mathbf{B}\boldsymbol{u}_t + \boldsymbol{v}_t, \qquad \boldsymbol{v}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{x}}), \tag{11}$$

$$\mathbf{A} = \mathbf{I}, \qquad \mathbf{B} = dt\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \boldsymbol{\Sigma}_{\boldsymbol{x}} = \begin{bmatrix} \sigma_{\text{rw}}^2 & 0 \\ 0 & \sigma_m^2 \end{bmatrix}. \tag{12}$$

The state cost penalizes the tracking error,

$$\mathbf{Q} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad \mathbf{R} = \begin{bmatrix} c \end{bmatrix}. \tag{13}$$

1.2.2 Point mass model

In the point mass model, the subject's hand follows a point-mass dynamical system with position, velocity, and force states, while the target still follows a random walk. Accordingly, the state vector is

$$\boldsymbol{x}_t = \begin{bmatrix} x_t^{\text{right}} \\ x_t^{\text{left}} \\ v_t^{\text{left}} \\ f_t^{\text{left}} \end{bmatrix}, \tag{14}$$

where the left-hand component is obtained from the continuous-time point-mass system (see e.g. Kasuga et al., 2022, for details)

$$m\dot{v}^{\text{left}} = -gv^{\text{left}} + f^{\text{left}}, \tag{15}$$

$$\tau\dot{f}^{\text{left}} = -f^{\text{left}} + u, \tag{16}$$

with damping g, mass m, and muscle time constant τ. Discretizing this linear system with time step dt yields the matrices $\mathbf{A}_{\text{pm}}$ and $\mathbf{B}_{\text{pm}}$; the corresponding process-noise covariance $\boldsymbol{\Sigma}_{\text{pm}}$ is obtained by discretizing the continuous-time noise model with van Loan's method (Van Loan, 2003). The resulting discrete-time dynamics matrices for the whole state are

$$\mathbf{A} = \operatorname{diag}\left(\mathbf{I}_1, \mathbf{A}_{\text{pm}}\right), \tag{17}$$

$$\mathbf{B} = \operatorname{diag}\left(0, \mathbf{B}_{\text{pm}}\right), \tag{18}$$

$$\boldsymbol{\Sigma}_{\boldsymbol{x}} = \operatorname{diag}\left(\sigma_{\text{rw}}^2, \boldsymbol{\Sigma}_{\text{pm}}\right). \tag{19}$$

The cost matrices are

$$\mathbf{Q} = \operatorname{diag}\left(\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, 0, 0\right), \tag{20}$$

$$\mathbf{R} = \begin{bmatrix} c \end{bmatrix}. \tag{21}$$

1.2.3 Observations and delay

We model delayed sensory observations as a time-stacked LQG system following Izawa and Shadmehr (2008) and Crevecoeur et al. (2016). The agent observes the signed difference between the right and left hand positions,

$$\boldsymbol{y}_t = \mathbf{H}\boldsymbol{x}_{t-\delta} + \boldsymbol{w}_t, \qquad \boldsymbol{w}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{y}}), \tag{22}$$

$$\mathbf{H} = \begin{bmatrix} 1 & -1 \end{bmatrix}, \qquad \boldsymbol{\Sigma}_{\boldsymbol{y}} = \begin{bmatrix} \sigma^2 \end{bmatrix}. \tag{23}$$

To handle the delay, we augment the latent state with the previous δ states,

$$\tilde{\boldsymbol{x}}_t = \left[\boldsymbol{x}_t, \boldsymbol{x}_{t-1}, \ldots, \boldsymbol{x}_{t-\delta}\right]^\top, \tag{24}$$

and use the corresponding block-companion dynamics and block-diagonal state cost,

$$\tilde{\boldsymbol{x}}_{t+1} = \tilde{\mathbf{A}}\tilde{\boldsymbol{x}}_t + \tilde{\mathbf{B}}\boldsymbol{u}_t + \tilde{\boldsymbol{v}}_t, \tag{25}$$

$$\tilde{\mathbf{Q}} = \operatorname{diag}\left([\mathbf{Q}, 0, \ldots, 0]\right). \tag{26}$$

This implementation of a delayed observation model is described in more detail by Izawa and Shadmehr (2008) and Crevecoeur et al. (2016). We assume $\delta = 2$ for vision and $\delta = 1$ for proprioception, which at $dt = 0.075s$ corresponds to a delay of 150 ms for vision and 75 ms for proprioception.

The point mass model with control cost and point mass model without control cost use the same delayed observation model, but differ in whether c is treated as a genuine penalty or set to a negligible value.

1.3 Model fitting

We fit the model to the behavioral data using inverse optimal control (IOC) for LQG system (Straub & Rothkopf, 2022). More concretely, IOC defines the likelihood of a trajectory of observed states conditional on the model parameters $p(\boldsymbol{x}_{1:T}^{(i)}|\boldsymbol{\theta})$, where $\boldsymbol{x}_{1:T}^{(i)}$ is the sequence containing T time steps of observed left and right hand positions in trial i. We assume that the individual trials are independent realizations of tracking behavior generated from the optimal control model with the same parameters $\boldsymbol{\theta}$, s.t. the likelihood of a dataset $\mathcal{D}$ is given by $p(\mathcal{D}|\boldsymbol{\theta}) = \prod_i p(\boldsymbol{x}_{1:T}^{(i)}|\boldsymbol{\theta})$.

We assume that the same control cost c and motor variability σ_m are applied in all conditions. We further assume that there is one proprioceptive noise parameter σ_{prop} that affects the proprioceptive conditions, and two visual noise parameters $\sigma_{\text{vis},1}$ and $\sigma_{\text{vis},2}$, one for each visual condition.

Overall, the model has four free parameters for each participant:

$$\boldsymbol{\theta} = \{c, \sigma_m, \sigma_{\text{prop}}, \sigma_{\text{vis},1}, \sigma_{\text{vis},2}\}.$$

We define weakly informative prior distributions over the parameters:

$$\begin{aligned} c &\sim \text{HalfNormal}(0.5) \\ \sigma_m &\sim \text{HalfNormal}(0.5) \\ \sigma_{\text{prop}} &\sim \text{HalfNormal}(20) \\ \sigma_{\text{vis},1} &\sim \text{HalfNormal}(20) \\ \sigma_{\text{vis},2} &\sim \text{HalfNormal}(20) \end{aligned}$$

To compare model assumptions, we consider the velocity control model and the point mass model, each with and without control cost. The difference between the control-cost-free and control-cost versions is whether c is fixed to a negligible value ($c = 10^{-6}$) or estimated as a free parameter.

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